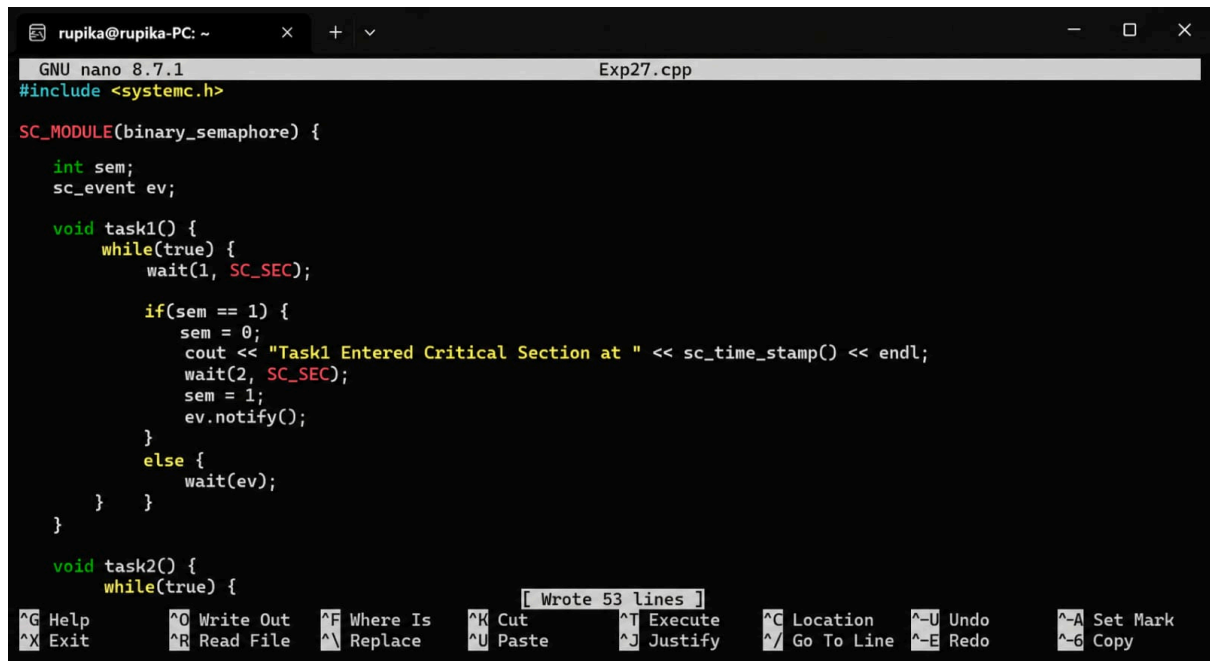


EXP:27



```
GNU nano 8.7.1 Exp27.cpp
#include <systemc.h>

SC_MODULE(binary_semaphore) {
    int sem;
    sc_event ev;

    void task1() {
        while(true) {
            wait(1, SC_SEC);

            if(sem == 1) {
                sem = 0;
                cout << "Task1 Entered Critical Section at " << sc_time_stamp() << endl;
                wait(2, SC_SEC);
                sem = 1;
                ev.notify();
            }
            else {
                wait(ev);
            }
        }
    }

    void task2() {
        while(true) {
```

[Wrote 53 lines]

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	^U Undo	^A Set Mark
^X Exit	^R Read File	^I Replace	^V Paste	^J Justify	^_ Go To Line	^E Redo	^6 Copy

```
rupika@rupika-PC:~$ nano Exp27.cpp
rupika@rupika-PC:~$ g++ -o Exp27 Exp27.cpp -lsystemc
rupika@rupika-PC:~$ ./Exp27

      SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
      Copyright (c) 1996-2025 by all Contributors,
      ALL RIGHTS RESERVED
Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s
rupika@rupika-PC:~$ |
```